PaddleOCR-VL-Circuit: Built for Schematic Diagram Understanding

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Abstract

Automatic recognition of circuit schematic diagrams is a core challenge in Electronic Design Automation (EDA), involving two heterogeneous tasks: visual text extraction and topological graph tracing. This paper presents CircuitOCR, a circuit schematic OCR and netlist extraction system based on PaddleOCR-VL-0.9B. We construct the first multi-source hybrid dataset, integrating 8,450 real-world open-source project schematics, 7,500 textbook circuit diagrams, and 14,000 procedurally generated synthetic renderings, totaling approximately 30,000 raw samples, of which 24,717 independent samples are retained after rigorous multi-round deduplication and quality screening. At the model level, we employ LoRA (rank $r = 8$, $\alpha = 16$) for parameter-efficient fine-tuning of PaddleOCR-VL-0.9B, updating only 0.30% of parameters (2.7M/908M) on a single NVIDIA RTX 4060 laptop GPU (8 GB VRAM). Constrained by 8 GB VRAM, Flash Attention is unavailable during inference, requiring a manual fallback implementation of attention computation, and max_length is limited to 30 tokens. On the easy50 test set, the original LoRA achieves Avg. NED = 0.8554 (+3.8%). After identifying the vision-language projection layer (Projector) as the key bottleneck through controlled experiments (E1–E4), adding Projector to LoRA targets further reduces Avg. NED to 0.8271 (+7.0%). 3-epoch training further lowers it to 0.8197 (+7.8%). Increasing the LoRA rank from $r = 8$ to $r = 16$ achieves the best Avg. NED = 0.8044 on easy50 (+9.6%), with 0.8291 on easy100 and 0.8624 on easy200, validating the gain from larger rank capacity in mitigating class collapse. The current absolute score is constrained by laptop GPU compute; this paper contributes primarily methodological validation and bottleneck analysis.

Keywords Circuit Schematic · OCR · PaddleOCR-VL · LoRA Fine-tuning · Netlist Extraction · Post-training · Low-VRAM Training

1 Introduction

Vision-Language Models (VLMs) have substantially expanded the application boundaries of traditional large language models, demonstrating significant advantages in scenarios requiring visual understanding, such as robotic navigation and document parsing tasks. Compared to traditional OCR pipelines that rely on multi-stage processing and substantial memory overhead, VLMs can directly output structured text from images in an end-to-end manner, eliminating cumbersome intermediate processing steps.

2 Background and System Setup

2.1 Base Model Specification

This project adopts PaddleOCR-VL as the open-source base model [Cui et al., 2025], whose underlying OCR capabilities are built upon the PP-OCR series of lightweight text detection and recognition systems [Li et al., 2022]. PaddleOCR-VL-0.9B is a Vision-Language Model (VLM) with 908M parameters, integrating a NaViT-style dynamic high-resolution vision encoder with the ERNIE-4.5-0.3B language model. It supports document parsing across 109 languages, covering common document elements such as text, tables, formulas, and charts. Its compact model size (0.9B) makes it suitable for fine-tuning and deployment on consumer-grade GPUs.

2.2 Engineering Scenario Definition

An Electronic Schematic Diagram is a two-dimensional engineering drawing that uses standardized graphical symbols to represent electronic components and their electrical connections. It reflects logical topology rather than physical layout. Circuit schematics belong to the EDA vertical industrial domain and serve as core carriers of engineering drawings and technical documentation; their automatic recognition and parsing hold significant engineering value. As highly complex technical documents, circuit schematics contain: text elements, including component reference designators (e.g., R1, C1), values (e.g., 10k, 100nF), net labels (e.g., VCC, GND, SPI-CLK), and title block text in the lower-right corner; special characters, including electronics-specific physical units and symbols such as Ω (ohm), μ (micro), and $\pm$ (tolerance); tabular elements, including multi-pin chip pinout tables, Bill of Materials (BOM), and revision change tables; and structured data, including circuit connectivity and network topology, requiring extraction of structured netlists analogous to formulas or code.

2.3 Limitations of Vanilla LLMs

The limitation of this approach stems from the severe deficiencies of the native, unfine-tuned base model in recognizing circuit schematic diagrams within technical documents. In the circuit schematic task, PaddleOCR-VL-0.9B fails to perform effective OCR recognition, exhibiting three typical failure modes: (1) outputting overlapping, repetitive word concatenations (e.g., repeatedly generating the same token); (2) generating generic text memorized from training data rather than performing recognition based on the input image (i.e., "hallucinating" rather than reading); and (3) occasionally outputting partially correct component labels while omitting the primary component names.

In the unfine-tuned setting, we conducted circuit schematic benchmark testing, where the base model's output consisted of meaningless format markers. When presented with circuit diagram inputs, the model either produced worthless garbled text, generated generic template responses, or fell into a degenerate mode of repeated token generation.

3 Dataset Construction

3.1 Raw Data Collection and Sources

We constructed the first high-quality and effective multi-source dataset [Zhang, 2026]. One portion of our dataset comes from the Open Schematic large-scale open-source dataset [bshada, 2025], comprising 8,450 real-world schematic diagrams and standard components from actual projects, totaling approximately 0.7 GB. This dataset is released under the original CC-BY-4.0 attribution license, permitting free open-source use and commercial applications. It adheres to real-world naming standards and conventions, with annotations in their own coordinate system independent of image pixel coordinates. The estimated Bounding Box accuracy of its annotations is approximately 71.4% (with some PCB occlusion noise). This dataset serves as the core training set for our model, suitable for training text recognition and netlist extraction models as a large-sample training corpus.

The second portion is the Masala-CHAI open-source dataset [Bhandari et al., 2025], also using the CC-BY-4.0 attribution license, with accompanying code partially under Apache-2.0. This dataset comprises circuit diagrams collected from electronic textbooks and teaching materials, with each image directly paired with a standard SPICE netlist ground truth. It directly aligns with the "schematic-to-netlist" task and is well-suited as a supervised fine-tuning (SFT) dataset for end-to-end image-to-SPICE netlist generation.

The third portion is our synthetic dataset [Zhang and Chen, 2026]. This dataset is procedurally generated via Python scripts, totaling approximately 14,000 rendered schematic images. The generation pipeline proceeds as follows: the program randomly generates circuit topology specifications (containing complex connection relationships with tens to hundreds of components), automatically draws the schematic and renders it as a PNG image, and simultaneously extracts annotation information directly from the circuit logic—including component reference designators (e.g., R1, C1), parameter values (e.g., 10k, 100nF), net labels (e.g., VCC, GND), and complete SPICE netlists, with 100% annotation accuracy. Circuit types span analog, digital, mixed-signal, and power management categories, with the component library continuously expanded across iterations to cover five general-purpose libraries. To simulate real-world image degradation, each schematic undergoes five types of degradation augmentation: paper aging (yellowing, speckling), scan noise and streaking, photographic perspective distortion, handwriting overlay, and low-resolution scanning. The synthetic dataset provides perfect ground truth and clean geometric boundaries, effectively compensating for the annotation accuracy and degradation coverage shortcomings of the two aforementioned open-source datasets.

Across the three data sources: Open Schematics and Masala-CHAI provide diverse real-world topologies and a vast array of real chip names (e.g., ATMEGA328P, TPS5430), while our synthetic dataset provides clean geometric boundaries and perfect ground truth. The complementarity of multi-source data enables the model to develop both broad familiarity with real component symbols and high pixel-level localization precision, preventing the model from recognizing only the font libraries or layouts of a single drawing tool.

3.2 Filtering, Deduplication and Quality Control

Open-source dataset overview

1. Open Schematics: approximately 8,450 images, real-world physical schematics, large-scale coverage of diverse real-world components.
2. Masala-CHAI: approximately 7,500 images, component-annotated diagrams with standard SPICE simulation netlists.

Synthetic dataset: 23 iterations, 35 types of perspective noise applied, totaling 14,000 images. Overall scale exceeds 30k samples.

3.2.1 Data Cleaning and Deduplication

To ensure the quality of data fed into the model, we constructed a pipeline of "Raw Data $\rightarrow$ Structural Filtering $\rightarrow$ Deduplication Module $\rightarrow$ Manual Re-inspection."

1. Deduplication Rules

Due to overlapping synthetic mapping in the image pixel libraries, duplicate schematics exist. We employ three deduplication strategies:

- **Text hash deduplication (literal):** Compute MD5/SHA256 hashes from text rendered as single PNG images from original, KiCad, and EDA sources, directly eliminating samples with identical text.
- **Topology hash deduplication (logical):** Different individuals drawing the same circuit may place components differently while maintaining identical electrical topology. We normalize and reorder component types and connection relationships, generate a topological feature string, compute its MD5 hash, and eliminate samples that are 100% logically identical.
- **Visual perceptual hashing (pHash/dHash):** Use perceptual hashing algorithms to compare entire PNG images, eliminating duplicate images with only minimal visual differences (e.g., only the version number or date in the bottom-right corner has changed).

2. Sample Quality Rules

- **Multi-component filtering:** Remove schematics with fewer than three components (many are lab prototypes, bare PCB block diagrams, or single package libraries).
- **Boundary and overlapping text cleaning:** Check whether annotation Bounding Boxes exceed image boundaries ($x < 0$, width overflow, etc.); for text box pairs with IoU > 0.8, where severe overlap renders them indistinguishable, remove the sample.
- **Distorted schematic culling:** Remove schematics containing internal hidden circuit symbol annotations such as KiCad `pwr` symbols (adding redundancy for such non-visible elements would induce model hallucinations).

Quality Inspection Pipeline

3.2.2 LLM-Based Quality Scoring and Purification

A vision large language model is introduced to automatically score all schematic images on a 1–5 scale across the following dimensions:

- **Image clarity:** Whether the text in the schematic is discernible to the human eye; eliminate excessively blurred or low-contrast degraded images.
- **Component density:** Whether the image is overly dense to the point of being indiscernible; whether the schematic is overcrowded.
- **Quality threshold:** Retain only samples with a composite score ≥ 3.

Manual Inspection For samples destined for evaluation and benchmark test sets, mandatory manual bounding-box verification is performed. A simple Gradio-based visual quality inspection tool was built: the image and predicted Bounding Boxes are rendered on the image, with the corresponding netlist displayed on the right side.

Manual Re-inspection and Review Samples missed during the initial screening by the quality inspection tool are manually verified and corrected. Rollback mechanism: systematic and widespread netlist logic errors discovered during large-scale inspection trigger a rollback modification of our synthetic generation code, followed by re-rendering of the affected data portion.

Execution of Cleaning

Data Cleaning and Multi-Source Quality Control Statistics The cleaning module simultaneously applied multiple rounds of deduplication and quality screening across Masala-CHAI, Open Schematics, and the synthetic dataset, totaling over 40,000 samples:

1. Initial screened samples: 39,947
2. After topology deduplication: 14,897 remaining
3. After visual hash and perceptual hash filtering: 9,117 (filtered out ∼7,000 duplicate schematic cases with identical topology but slight image modifications); logical deduplication: 1,580 (removed samples with identical topology but duplicate naming parameters and annotations); SFT sample automatic screening: 10,240 samples, 1,100 duplicates removed; visual scoring evaluation: 2,530 (filtered out extremely low-clarity, minutely blurred and distorted real-world schematics via LLM evaluation; 0 anomalous samples scoring below 3 were removed, i.e., all samples in this batch passed quality screening).

Dataset Splitting and PaddleOCR-VL SFT Format Conversion After the rigorous cleaning described above, a total of 24,717 independent schematic samples were obtained. These were split into training, validation, and test sets following a standard 70%/15%/15% ratio: 17,302 training samples, 3,708 validation samples, and 3,707 test samples. Three rounds of naming integrity verification were conducted, confirming compliance of all resulting JSON files. After SFT format conversion, the actual training set used for this training run consists of 2,433 samples, with 523 samples in the test set.

Note: Full manual verification was not performed. Due to time and manpower constraints on one hand, and the inherent rigor of the automated data cleaning pipeline and multi-source dataset conventions on the other, the correctness of the dataset has been ensured to the greatest extent possible.

Dataset Stratification

- **Synthetic dataset:** Drawn directly within the program by the circuit generation algorithm, requiring no annotation (bbox); connection relationships are directly output by program logic, achieving absolute precision in annotated characters and multi-value information. Masala-CHAI and Open Schematics samples originate from textbooks and real-world GitHub project schematics, with inherent annotation discrepancies.
- **Triple-filter cleaning pipeline:** For real-world schematics that may contain noise and imperfections, we adopted rigorous automated verification and employed LLM-based visual self-inspection, using AI to replace manual preliminary review.
- **Output format consistency verification:** Line-by-line validation was performed on the split samples, achieving 100% alignment.

3.3 Data Distribution and Statistics

After cleaning, the final dataset contains 24,717 independent schematic samples. The dataset is split 70%/15%/15% into training (17,302 samples), validation (3,708 samples), and test (3,707 samples) sets. The validation and test sets are kept at reasonable sizes to ensure feasibility of manual verification, while the training set provides sufficient fine-tuning data. Table 1 lists key statistics for each data source.

Label Length Distribution. The training set exhibits an extremely wide span of label text lengths, ranging from short component lists (5–10 tokens) to complete SPICE netlists (200+ tokens), reflecting the diversity of the circuit schematic OCR task. Based on label length, we partition the test set into four graded difficulty tiers: easy50 (label length 27–109 characters), easy100 (27–163 characters), easy200 (27–296 characters), and full523 (all 523 samples), to systematically evaluate model performance across different complexity levels.

Table 1: Dataset Composition and Source Statistics

Source	Samples	License	Characteristics
Open Schematics	∼8,450	CC-BY-4.0	Real open-source hardware KiCad schematics
Masala-CHAI	∼7,500	CC-BY-4.0	Textbook circuits with SPICE netlist GT
Synthetic	∼14,000	Apache 2.0	Procedurally generated, 100% accurate annotations
Total	∼30,000	–	Analog/Digital/Mixed-signal/Power categories

Circuit Type Coverage. The dataset covers four major circuit categories: analog circuits (op-amps, filters, power management), digital circuits (logic gates, MCU peripherals, memory interfaces), mixed-signal circuits (ADC/DAC, PLL), and power circuits (DC-DC converters, LDO regulators). The procedural generation of the synthetic dataset ensures adequate training samples for each category.

3.4 Data Augmentation Strategy

During training, we apply the following online augmentations to each input image: random JPEG compression (quality factor 85–100), random scaling (max dimension 168–224 pixels), and random horizontal flipping. These lightweight augmentations simulate real-world image quality variations while preserving the readability of the circuit schematics.

4 Model Fine-tuning and Performance Improvement Roadmap

4.1 Model Fine-tuning

We employ LoRA (Low-Rank Adaptation) [Hu et al., 2021] for parameter-efficient fine-tuning of PaddleOCR-VL-0.9B.

4.2 Performance Improvement Roadmap: from 3.8% to 9.6% Validated

The original LoRA fine-tuning achieves a 3.8% Avg. NED improvement (0.8895 $\rightarrow$ 0.8554). After identifying the Projector bottleneck through E1–E4 experiments, adding Projector LoRA brought the improvement to 7.0% (0.8271). 3-epoch training achieved 7.8% (0.8197). Increasing LoRA rank to $r = 16$ further achieved 9.6% (0.8044). Below we summarize the identified bottlenecks, validated improvements, and future directions.

4.2.1 Identified Bottlenecks

1. **Projector frozen (Core bottleneck).** PaddleOCR-VL's vision-language projection layer (`mlp_AR`, 25.9M parameters) is responsible for mapping image features output by the vision encoder to the language model's embedding space. The current LoRA only covers the q/k/v/o projections of self-attention layers; the Projector's `linear_1` (4608$\times$4608) and `linear_2` (4608$\times$1024) are completely frozen. This means the mapping from visual features to language space remains unchanged throughout training, preventing the model from learning visual-semantic correspondences of circuit symbols. Experiment E4 confirmed this bottleneck, but adding Projector LoRA caused OOM under the current 8 GB VRAM constraint.
2. **Insufficient training.** Initial experiments used only 1 epoch, max_dim=168px, batch_size=1. Loss decreased from 3.0 to 1.2 but had not yet converged, indicating substantial room for optimization. The class collapse at 100 samples and partial mitigation at 2,433 samples demonstrated that data volume is critical for maintaining output diversity, but 2,433 samples may still be insufficient to cover the full vocabulary distribution of circuit OCR. Subsequent 3-epoch training and $r = 16$ rank expansion have partially alleviated this issue.
3. **Inferenceconstraints.** Flash Attention is unavailable on Windows + RTX 4060, requiring manual fallback implementation. The 8 GB VRAM limits max_length to 30 tokens, far insufficient for outputting complete SPICE netlists (typically requiring 100–500 tokens). The 47/50 sample completion rate indicates the currentapproaches isnear the VRAM limit.
4. **Limitations of pure text NED as the sole metric.** NED cannot distinguish between circuit semantic correctness and character-level similarity, nor reflect the correctness of topological connections. The topology evaluation framework (component F1, pin accuracy) has been designed but not yet implemented.

4.2.2 Improvement Plan (Mostly Validated on Current Hardware)

Step 1: Unlock the Projector bottleneck (Validated). The Projector's LoRA parameter count is only approximately 120K (`linear_1`+`linear_2`), and will not significantly increase VRAM during training. The OOM in Experiment E4 occurred during inference (full attention computation during generation), not during training. Therefore, Projector LoRA can be added during training, with inferenceverification by further reducing max_length (e.g., 15–20 tokens) or image dimensions. If Projector LoRA yields a substantial NED improvement (e.g., $> 1\%$), it will prove that this bottleneck is worth investing in on GPUs with larger VRAM.

Step 2: Multi-epoch training (Validated). Increasing epochs from 1 to 3 (7,299 steps, 77 min) on top of Projector LoRA achieved Avg. NED = 0.8197 on easy50, a further 0.9% improvement over 1-epoch (0.8271) and a cumulative 7.8% over baseline. Diminishing returns suggest limited marginal benefit from further epoch increases, but multi-epoch training still contributes positively to mitigating class collapse.

Step 2.5: Larger LoRA rank (Validated). Increasing the LoRA rank from $r = 8$ to $r = 16$ ($\alpha = 32$, 5.7M trainable parameters) with 3-epoch Projector LoRA training took 53 min. The $r = 16$ configuration achieved the best Avg. NED = 0.8044 on easy50 (cumulative +9.6% vs. baseline), with 0.8291 on easy100 and 0.8624 on easy200. Compared to $r = 8$ (easy50 NED 0.8197), $r = 16$ provides an additional 1.9% improvement, validating the positive contribution of larger capacity to mitigating class collapse.

Step 3: Diverse decoding strategies (Attempted). Top-p/top-k sampling with temperature tuning ($T = 1.0 \sim 1.5$) was tested on both $r = 8$ and $r = 16$ models. However, due to severe output distribution collapse after LoRA fine-tuning (most probability mass concentrated on a few tokens), sampling failed to effectively increase output diversity, with outputs still showing repetitive tokens. Improving decoding strategies may require first addressing the distribution collapse issue.

Step 4: Implement topology evaluation. Encode the designed topology evaluation protocol (component F1, type accuracy, pin accuracy) as automated scripts. Since NED has been shown to be ineffective at distinguishing circuit outputs of different quality (base model garbled NED=0.89 vs. LoRA circuit-label NED=0.92), topology evaluation can provide more meaningful feedback signals for model improvement.

4.2.3 Long-term Directions (Requiring Additional Resources)

If the near-term improvements raise NED to a practically useful level (e.g., < 0.5), the following directions are worth further exploration:

- **Multi-task decoupling:** Decompose circuit OCR into two independent sub-tasks of text recognition and topology tracing, separately fine-tuning LoRA adapters responsible for different regions, and merging them at inference via LoRA switching to achieve multi-expert fusion-like effects at low computational overhead.
- **Simulator-based verification reward:** Integrate Ngspice/LTspice simulation into the training loop, using the electrical correctness of netlists as an additional reward signal. This requires deploying a simulator in the training environment and automatically capturing simulation output.
- **Larger base models:** Transfer the current LoRA fine-tuning approach to larger-scale vision-language models in the PaddlePaddle ecosystem, exploring the impact of model capacity on circuit schematic understanding tasks. Larger vision encoders can extract finer-grained circuit symbol features.

Among the above approaches, Steps 1/2/2.5 have been validated (cumulative +9.6%), Step 3 was attempted but showed limited effect due to distribution collapse, and Step 4 remains to be implemented. The long-term directions require additional computational and engineering resources and are discussed here only as directional guidance.

5 Experimental Evaluation

5.1 Experimental Setup

All experiments were conducted on a single NVIDIA RTX 4060 (8 GB VRAM). The model uses PaddleOCR-VL-0.9B as the base, with LoRA fine-tuning configured at rank $r = 8$, $\alpha = 16$, targeting the q_proj, k_proj, v_proj, and o_proj projection matrices of all attention layers. Training employs the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight_decay=0.1), learning rate 5×10^{-4}, cosine annealing to 5×10^{-5}, 1 epoch, maximum image dimension 168 pixels. Total trainable parameters amount to 2,746,368, representing 0.30% of the total model parameters (908M).

The evaluation metric is Average Normalized Edit Distance (Avg. NED), defined as:

$$\text{NED}(p, g) = \frac{\text{LevenshteinDistance}(p, g)}{\max(|p|, |g|)} \tag{1}$$

where p is the model prediction text and g is the ground truth annotation. NED = 0 indicates a perfect match, and NED = 1 indicates a complete mismatch.

5.2 Baseline Benchmark Results

The unfine-tuned PaddleOCR-VL-0.9B base model performs extremely poorly on the circuit schematic OCR task. Its baseline scores across test tiers are: easy50 NED = 0.8895, easy100 NED = 0.8999, easy200 NED = 0.9139. Table 2 shows the detailed comparison on the easy50 tier. Base model outputs primarily exhibit three failure modes: (1) degenerate repetition of a single token (e.g., continuously outputting "GND"); (2) generation of generic text memorized from training data rather than image-conditioned recognition (e.g., outputting Arduino tutorial fragments); (3) outputting partially circuit-relevant symbols unrelated to the input image content.

Table 2: easy50 Benchmark Full Comparison (max_length=30)

Configuration	Train Samples	Train Time	Valid Samples	Avg. NED	vs. Base
Base (unfine-tuned)	0	–	50/50	0.8895	–
LoRA $r = 8$ (q/k/v/o)	2,433	100 min	47/50	0.8554	+3.8%
LoRA $r = 8$ (+Projector)	2,433	27 min	53/50	0.8271	+7.0%
LoRA $r = 8$ (+Projector, 3 ep)	2,433	77 min	50/50	0.8197	+7.8%
LoRA $r = 16$ (+Projector, 3 ep)	2,433	53 min	50/50	0.8044	**+9.6%**

Note: (1) 3 samples failed (OOM) due to insufficient GPU VRAM (8 GB) during Flash Attention fallback computation.
(2) Lower NED is better; NED=0 = perfect match, NED=1 = complete mismatch.

5.3 LoRA Fine-tuning Experimental Results

After full training on 2,433 samples, original LoRA (q/k/v/o) achieved Avg. NED 0.8554 (+3.8%, 47/50 samples). Adding Projector LoRA improved it to 0.8271 (+7.0%, 50/50). 3-epoch training achieved 0.8197 (+7.8%, 50/50). Increasing rank to $r = 16$ further reduced NED to 0.8044 (+9.6%, 50/50), validating the contribution of larger rank capacity. Complete results in Table 2.

5.4 Bottleneck Diagnostic Experiments

To identify the key bottlenecks limiting LoRA fine-tuning effectiveness, we designed four controlled variable experiments (E1–E4).

5.5 Topology Evaluation (Design Stage)

In addition to text-level NED metrics, we have designed a topology-level evaluation protocol. Model outputs are parsed into structured netlists, from which three dimensions of topology metrics are extracted:

- **Component F1 Score:** Compares predicted component designators (e.g., R1, C3) against the ground truth component set for exact matching.
- **Type Accuracy:** Evaluates the correctness of component type classification (resistor, capacitor, MOSFET, etc.).
- **Pin Connection Accuracy:** Evaluates the proportion of component pins correctly connected to their corresponding nets.

Topology evaluation provides a more fine-grained measure of circuit understanding capability than pure text NED, capable of distinguishing whether the model genuinely comprehends electrical connectivity relationships or merely matches at the text level.

Table 3: Bottleneck Diagnostic Experiment Design and Results

Experiment	Variable	Result	Conclusion
E1: Blank image	Real vs. solid gray	Output differs	Vision encoder works; model “sees” images
E2: Resolution	max_dim 168 $\to$ 336	Still collapsed	Resolution is not root cause
E3: Epoch count	1 vs. 3 epochs	Validated (3-epoch NED 0.8197)	Multi-epoch has positive gain
E4: LoRA layers	Add Projector layers	Validated (NED 0.8271, +7.0%)	Projector is key bottleneck
E5: LoRA rank	$r = 8 \to r = 16$	Validated (NED 0.8044, +9.6%)	Larger rank further improves performance

Key findings:
(1) E1 confirms the LoRA fine-tuned model’s vision capability is functional and output varies with images;
(2) E2 rules out the image resolution hypothesis;
(3) E4 identifies the Projector (mlp_AR, 25.9M parameters) as the sole bridge for vision-language mapping being frozen as the root cause of insufficient training, but adding Projector LoRA causes VRAM overflow.

Table 4: Training Hyperparameters and System Configuration

Parameter	Value
Base model	PaddleOCR-VL-0.9B (908M parameters)
LoRA config	$r = 8$, $\alpha = 16$, dropout=0.0
LoRA target modules	q_proj, k_proj, v_proj, o_proj (153 pairs)
Trainable params	2,746,368 (0.30% of total)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, weight_decay=0.1)
LR schedule	Cosine annealing $5 \times 10^{-4} \to 5 \times 10^{-5}$
Training epochs	1
Max image dim	168 px (constrained by 8 GB VRAM)
Label truncation	200 characters
Inference max_length	30 tokens (constrained by Flash Attention unavailability)
GPU	NVIDIA RTX 4060 Laptop (8 GB VRAM, CUDA 12.6, cuDNN 9.5)
OS	Windows 11, PaddlePaddle 2.6.2

6 Empirical Analysis

6.1 Base Model Failure Mode Analysis

We conducted qualitative analysis of the base model’s outputs on circuit schematics andsummarized three typical failure modes:

1. **Token Repetition Degeneration:** The model becomes trapped in aloop of repeatedly outputting the same token. For example, in response to a schematic containing multiple component designators and netlist lines, the model outputs a continuous repetition of “GND GND GND GND …” This suggests that the model loses autoregressive decoding diversity when faced with out-of-domain images, collapsing to a degenerate distribution.
2. **Memory Hallucination:** The model does not perform image-conditioned generation but instead outputs generic text fragments memorized from its pre-training data (e.g., webpage navigation labels, Markdown formatting markers). This indicates that while thepre-trained PaddleOCR-VL possesses strong general document understanding capabilities, it has not established the visual-semantic mapping between circuit symbols and their corresponding text.
3. **Partial Recognition and Information Omission:** The model occasionally correctly identifies individual component designators (e.g., “R1”) but omits the main component names, net labels, and parameter values present in the drawing. The output length is far shorter than the ground truth annotation, suggesting an attention distribution problem across the circuit diagram layout.

The root cause of these failure modes lies in the fact that although PaddleOCR-VL performs excellently on multilingual document parsing, its pre-training data predominantly consists of general documents (invoices, forms, book pages) and rarely includes highly specialized engineering drawings such as circuit schematics. The visual features of circuit symbols, Greek letter units, and electrical connection lines differ significantly from conventional document elements.

6.2 Effect of Fine-tuning on Output—Positive Effects and Current Limitations

After 100-sample LoRA fine-tuning, the model's output behavior underwent the following changes:

Positive Effects:

- The domainaffiliation of output content underwent a fundamental shift: from the base model's generic text (Arduino tutorial fragments, Markdown markers) to circuit-domain labels (component values "1.2 V," net labels "GND"). This demonstrates that LoRA fine-tuning successfully established the visual-semantic mapping between circuit symbols and their corresponding text.
- Output length increased from the base model's extreme brevity (2–5 characters) to lengths comparable to circuit annotations.

Diagnostic finding: Class collapse and NED limitations.

In the 100-sample quick verification, we observed a diagnostic phenomenon: the model achieved an NED of only 0.9184. One hundred samples (only 4.1% of the full 2,433 training set) are insufficient for the model to learn adequate lexical diversity; the model learned the highest-frequency circuit label `GND` in the training set and then over-relied on this label, degenerating into single-label repetition during testing. This phenomenon reveals the risk of Class Collapse—few-sample LoRA fine-tuning can cause severe contraction of the model's output distribution, akin to catastrophic forgetting. Meanwhile, the base model'sgarbled output achieves a paradoxically lower (better) NED (0.8895) due to character diversity, demonstrating that pure text NED cannot distinguish between "domain-relevant but monotonous output" and "domain-irrelevant but diversegarbled output," necessitating topology-level evaluation metrics as a supplement.

Post-full-training status.

Based on the Projector bottleneck identified in E4, we added the vision-language projection layers (mlp_AR.linear_1 and linear_2, ≈120K extra parameters) to the LoRA targets. This configuration achieved Avg. NED = 0.8271 on easy50, a further 3.3% improvement over the original LoRA (0.8554) and a cumulative 7.0% over the baseline (0.8895). This directly validates the E4 diagnosis—the Projector, as the critical bridge mapping visual features to language space, is a key bottleneck when frozen.

After 3-epoch Projector LoRA training, the NED further decreased from 0.8271 to 0.8197 (cumulative +7.8% vs. baseline), confirming the gain from multi-epoch training.

Effect of larger LoRA rank. Increasing the LoRA rank from $r = 8$ to $r = 16$ ($\alpha = 32$), 3-epoch Projector LoRA training took 53 min. The $r = 16$ configuration achieved the best Avg. NED = 0.8044 on easy50 (+9.6% vs. base 0.8895), 0.8291 on easy100 (+7.9% vs. base 0.8999), and 0.8624 on easy200 (+5.6% vs. base 0.9139). Compared to $r = 8$ (easy50 NED 0.8197), $r = 16$ provides an additional 1.9% improvement, and training time is shorter (53 vs. 77 min), suggesting that larger rank may accelerate convergence. The diminishing improvement across tiers is expected as circuit complexity increases. Cross-tier performance is shown in Table 5.

Table 5: $r = 16$ Projector LoRA vs. base model across test tiers

Test Tier	Samples	Base NED	$r = 16$ NED	Improvement
easy50	50	0.8895	0.8044	+9.6%
easy100	100	0.8999	0.8291	+7.9%
easy200	200	0.9139	0.8624	+5.6%

The optimal NED of 0.8044 remains far from practical usability (NED < 0.5), severely constrained by laptop GPU compute (see Section 6.3). On larger GPUs, Flash Attention and higher max_length/max_dim are expected to yield significantly better absolute scores.

6.3 Dataset Contribution

The multi-source hybrid dataset construction strategy has proven effective. Open Schematics provides real-world component naming diversity (e.g., specific chip models such as ATMEGA328P and TPS5430), Masala-CHAI provides textbook-grade standard SPICE netlist formats, and the synthetic dataset provides precise annotations and controlled degradation scenarios. The complementarity of the three sources eliminates the overfitting risk of any single source, enabling the model to develop both generalization ability to real component symbols and strict adherence to annotation accuracy.

7 Conclusion and Limitations

7.1 Major Contributions and Current Progress

This paper presents CircuitOCR, a complete solution for circuit schematic OCR and netlist extraction. Current work progress is as follows:

1. **Multi-source hybrid dataset (Completed):** Constructed the first large-scale circuit schematic dataset integrating real open-source projects, textbook circuits, and procedurally generated synthetic data. After three rounds of deduplication (text hash, topology hash, visual perceptual hash) and LLM-based quality scoring, approximately 24,717 independent samples were retained.
2. **LoRA fine-tuning scheme (Completed):** Original LoRA (q/k/v/o) achieved Avg. NED 0.8554 (+3.8%) on easy50. Adding Projector LoRA further reduced NED to 0.8271 (+7.0%). 3-epoch training achieved 0.8197 (+7.8%). Increasing rank to $r = 16$ achieved the following across three test tiers: easy50 NED 0.8044 (base 0.8895, +9.6%), easy100 NED 0.8291 (base 0.8999, +7.9%), easy200 NED 0.8624 (base 0.9139, +5.6%). Class collapse at low sample counts was mitigated by full training and larger rank.
3. **Performance improvement roadmap (Planning stage):** Based on diagnostic experiment conclusions, four near-term actionable improvement directions are planned: Projector LoRA, multi-epoch training, diverse decoding, and topology evaluation; as well as three long-term exploration directions: multi-task decoupling, simulation verification, and larger base models.
4. **Multi-dimensional evaluation framework (Partially completed):** Established a dual-layer protocol of text NED and topology evaluation, but the full implementation of topology evaluation (component F1, pin accuracy) remains to be completed.

7.2 Identified Problems and Lessons

1. **Class collapse risk and NED limitations:** Few-sample LoRA fine-tuning carries a class collapse risk—the model over-relies on a few high-frequency labels, leading to loss of output diversity. Furthermore, pure text NED has limitations in evaluating circuit OCR quality: non-domain-relevant random output may obtain a better NED score due to character diversity. See Section 5 for details.
2. **Engineering challenges:** Under the Windows + PaddlePaddle 2.6.2 environment, GPU bfloat16 support has known limitations: flash_attention is unavailable and requires manual fallback implementation; PaddleFormers' fp8 dtype management mechanism has compatibility conflicts with PaddlePaddle's native fp8 (requiring removal of global float8 alias to ensure correct model weight read/write). These engineering issues consumed substantial debugging time and reminds us that the community needs to strengthen E2E training stability testing on the Windows platform.
3. **Prerequisites for post-training:** The engineering infrastructure required for long-term directions such as multi-task decoupling and simulator verification (Ngspice integration, multi-LoRA switching, larger GPU VRAM) far exceeds the capabilities of the current single-GPU laptop workstation.

7.3 Limitations

1. **Severe compute constraints:** All experiments were conducted on a single laptop RTX 4060 (8 GB VRAM), which severely constrained outcomes: (a) Flash Attention unavailable, requiring manual fallback with $\sim$10x slower inference; (b) max_length limited to 30 tokens, far insufficient for complete SPICE netlists (100–500 tokens); (c) max_dim limited to 168px, causing loss of fine text details; (d) batch_size=1 with no gradient accumulation; (e) Windows + PaddlePaddle 2.6.2 bfloat16 issues added debugging overhead. These constraints resulted in optimal NED of 0.8044 (+9.6%). On GPUs with larger VRAM, significantly better absolute scores are expected. This paper contributes primarily methodological validation and bottleneck analysis, not absolute scores.
2. **Computational resource constraints:** Experiments were conducted on a single RTX 4060 8 GB laptop GPU, significantly limiting batch size, training speed, and model scale.
3. **Post-training mostly complete:** Original LoRA, Projector LoRA, 3-epoch training, and $r = 16$ rank expansion have been completed (optimal NED 0.8044, +9.6%). Diverse decoding was attempted but showed limited effect due to distribution collapse. Topology evaluation and long-term directions such as multi-task decoupling and simulation verification remain to be implemented.
4. **Manual verification not performed:** Final manual review of the dataset was not completed.

7.4 Future Work

Future work will focus on: (1) benchmarking on the full523 test set; (2) implementing the topology evaluation protocol (component F1, pin accuracy); (3) exploring $r = 32$ and other higher ranks for marginal gains; (4) introducing regularization (e.g., label smoothing) during training to mitigate distribution collapse, enabling diverse decoding to be effective; (5) exploring long-term directions such as multi-task decoupling and simulator-based verification on GPUs with larger VRAM.

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